MIT 18.06SC-
Exercises 2.5

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Problem 19.1

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$|A| = 1\,(-(1 \times 1 \times 1))$$
$$= -1$$

Cofactor method is preferable because the ~~determinan~~ 3x3 matrix we multiply -1 with to get $|A|$ is diagonal thus making it very easy to compute. Row exchange method or using the big formula would take many more steps for the same answer.

Problem 19.2

We know that $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix}$ has determinant 1

Also, by cofactors, the determinant of the second matrix (i.e. the one with det 0) is a sum of products, where one of the products is the value of the nxn entry multiplied by 1 (i.e the nxn entry itself). When the nxn entry gets reduced by 1, and the sum

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before was 1~~0~~, it naturally leads to the new sum / determinant being 0.